

How do warmth and competence cues shape social networks?

An AMB project

Introduction :

In sociology, understanding how micro-level individual behaviours aggregate into macro-level social structures remains a fundamental puzzle in sociology. Analytical Sociology addresses this challenge by identifying the precise mechanisms, finding the simple and generalizable rules, that explain how individual actions produce observable social outcomes (Hedström, 2021). This approach adds an mechanism based explanation rather than simply documenting correlations. It demands causal explanations that are grounded in micro level processes.

Coleman's boat provides the conceptual framework for the micro-macro transition. A sociological explanation should be based on three level of explanations. In the first place, it needs to explain how existing structures shape individual actions (macro-micro). Secondly, how the individual behave based on their motivation and perception (micro-micro). And lastly, on how these individual aggregate actions leads to emergent macro-level outcomes (micro-macro) (Coleman, 1986).

Non-verbal behaviour is a fundamental yet often overlooked mechanism in social interaction. Perceived body language, facial expressions, eye contact, and posture operate as rapid, automatic communication channels that shape first impressions instantaneously and more often than we want to acknowledge it, dictates how an individual will behave. According to Fiske et al. (2007), these impressions universally organize around two independent dimensions that are : Warmth (W) and Competence (C). Through evolutionary basis, Warmth signals intent (friend or foe), while Competence signals capability (can that person act on that intent). Through emotion contagion, these subconscious evaluations trigger specific emotional responses : admiration, pity, envy, or contempt (as per the BIAS map), which guide approach or avoidance behaviours.

In this paper I am aiming to apply Analytical Sociology and Computational Social Science principles to examine how non-verbal warmth and competence cues translate into emergent social network structures. Using an Agent-Based Model (ABM), I will simulate how individual perceptions drive preferential tie formation, ultimately producing observable network patterns at the macro level. The central theoretical hypothesis claims that agents perceived as high on both dimensions (the "admiration" profile) will attract significantly more connections, becoming charismatic hubs, while those rated low on both will face systematic isolation (Cuddy, Fiske & Glick, 2008).

I will first establish the theoretical foundations, linking my model to a middle-range theory. Then I will present the ODD protocol detailing the model's mechanisms. After that comes the results and analysis from the factorial experiment, examining the parameter effects, the social exclusion, the hub formations, and the emotional category outcomes. Finally, I will discuss the

findings in relation to causal inference arguments for and against ABMs. And lastly, conclude with theoretical contributions and limitations my model may bring.

Literature review and Theoretical gap :

The Stereotype Content Model (SCM) and BIAS map from Fiske et al., 2007 and Cuddy et al., 2008, establishes that perceived warmth and competence evaluations shape behavioural tendencies. Admiration provokes approach whereas, contempt provokes avoidance. However, these frameworks remain descriptive, outlining individual-level processes without explaining how they scale up to produce meso or macro level network structures. Existing research rarely looks at how repeated warmth–competence evaluations, formed solely from non-verbal first impressions, build up over interactions and eventually shape network structures like clusters or isolates.

Furthermore, very little work explores whether warmth and competence combine in a multiplicative rather than additive way when shaping attraction, or whether people prefer more “balanced” profiles, where warmth and competence are displayed at a similar degree, which could influence how networks form (Fiske et al., 2007). And although evolutionary psychology suggests that warmth is assessed first (Keating, 1985), computational models that actually test these dynamics in network settings remain rare.

I aim to address these gaps by using an agent-based model and factorial experiment connecting theoretical insights to emergent network patterns. Each agents are assigned a continuous warmth and competence attributes that represent a set of non-verbal cues in the form of a single value. They will evaluate each another using an attraction function that includes: (1) a multiplicative interaction term ($W \times C$), (2) an imbalance penalty ($\beta |W - C|$), (3) emotional modifiers based on BIAS categories, and (4) a baseline openness parameter (τ).

This integrative design incorporates the SCM/BIAS framework, evolutionary theories on nonverbal communication, and Coleman's Boat to form a cohesive computing system. The model investigates whether predicted patterns, high status for high-warmth/high-competence individuals and marginalization of low-warmth/low-competence individuals, are sufficient, in the absence of verbal communication, to produce structured social networks with charismatic hubs and isolated actors.

Section 1 :

Theoretical Framework: ABMs as Tools for Middle-Range Theory

The main objective of Analytical Sociology (AS) is to provide robust, causal explanations for social phenomena by focusing on the underlying social mechanisms (Hedström, 2021). To do this, the methodological imperative needs to bridge the gap between individual actions (micro) and collective outcomes (macro). My Factorial Experiment (FE), an Agent-Based Model (ABM) examining the emergence of social networks from non-verbal cues, is fundamentally

rooted in this analytical framework, specifically serving as a tool for constructing and testing a middle-range theory (MRT) (Squazzoni, 2012).

1. ABMs as Tools for Middle-Range Theory

Middle-range theories, as originally conceptualized by Merton, focus on specific observable phenomena rather than grand universal laws. ABMs are perfectly suited for this purpose because they formalize the very mechanisms that constitute these theories, linking micro-level assumptions to macro-level predictions. Squazzoni (2012) explicitly positions ABMs within the "Analytical Models" category, emphasizing their capacity for revealing non-obvious properties of social systems **and** providing well-specified examples for middle-range theories (Squazzoni, 2012, p. 22).

Therefore, my FE aims to formalize a specific MRT wants to test how non-verbal-based evaluations of Warmth and Competence drive a differential approach/avoidance mechanism that generates structural inequality (hubs and isolates) in emergent social networks.

2. Critical Assessment: Theoretical vs. Empirical Realism

The FCP, like many analytical ABMs, comes with a trade-off between different types of realism (Manzo, 2022). In my case, I prioritize theoretical realism and intentionally accept lower input and output realism.

High Theoretical Realism: I derived the agent's behaviour rules (W/C, BIAS map, tie-formation logic) directly from empirically supported theories in social perception and evolutionary psychology (Keating, Antonakis). By using composite W/C variables based on non-verbal cues (following Heide's classification), I luckily avoid having to model every micro-behaviour separately, while still maintaining a psychological coherence.

Low Input/Output Realism: According to Manzo, the FCP is an exploratory, "theoretically informed" ABM (Manzo, 2022). I haven't used empirical data to calibrate the parameters (τ , β), and I also do not compare the simulated outcomes to real-world network data for validation, which reflects low Output Realism.

This simplification, and especially keeping W and C constant and excluding verbal communication, is a key strength for theory building. By isolating the non-verbal W/C mechanism, I can use my FE to test a strong hypothesis. Thus, if this single mechanism can generate structural inequality, then the model provides a minimal, well-specified explanation for part of social stratification. This isolation is crucial for making a clear mechanistic causal argument without getting lost in complex data calibration.

Methodology/ODD :

1. Overview

a. Purpose

The purpose of this agent-based model is to explore how non-verbal cues of warmth and competence influence first impressions and the resulting social network structure. The model examines why some individuals become socially central ("charismatic hubs"), while others remain peripheral or isolated, even in the absence of verbal communication.

Each agent emits two types of non-verbal signals: warmth cues (friendliness, openness, positive emotional expression) and competence cues (posture, confidence, controlled expressiveness). Together, these define the agent's perceived attractiveness for social connection.

The model aims to:

1. Identify the combinations of warmth and competence that produce high centrality
2. Observe whether clusters form around highly attractive agents
3. Determine conditions under which agents become neglected or isolated
4. Test how trait imbalance and baseline social openness affect network formation

This approach builds conceptually on the CoMSES model "The effect of homophily on co-offending outcomes" (2023), but replaces homophily with attraction evaluation based on warmth $\times$ competence interactions.

Specifically, the goal is to identify which levels of warmth and competence make an agent likely to be chosen as an interaction partner (network centrality) versus ignored (isolation). Verbal communication is explicitly absent and the whole group has different W and C values.

b. State Variables and Scales

Entities include:

- **Agents** (of any gender, age, SES, etc.)
- **Directed links** (social ties representing "willingness to connect with this agent" / intention to interact)

Agent-level attributes:

Each agent i has:

- **warmth_i $\in [0,1]$** : Intensity of warmth non-verbal cues (smiling, open posture, friendly gaze, etc.). Warmth cues represent a combined set of proven warmth signals in the form of a single value.

- **competence_i $\in [0,1]$** : Intensity of competence/power cues (good posture, controlled gestures, taking space, etc.). Competence cues represent a set of proven competence signals in the form of a single value.
- **num-links_i**: Number of outgoing ties created (maximum = 5)

Warmth and competence values are drawn randomly (uniform distribution) at initialization and remain constant during the simulation (no learning, no updating of traits).

Link attributes:

- Links are binary (present or absent)
- A directed link $i \rightarrow j$ means: "agent i chooses j as a desired interaction partner"
- Maximum of 5 outgoing links per agent (MAX-OUTGOING-LINKS = 5)
- Links accumulate over time and are not removed

Temporal and Spatial Scales:

- 1 tick = 1 round of first impressions
- Total simulation steps: 30 ticks
- Agents can perceive every other agent at every tick
- Spatial distance has no effect on visibility
- World size is arbitrary (used for display only)

Environment:

A directed link from agent i to agent j represents that i would like to interact with j based on first impressions. There is no explicit physical environment; the model focuses on the social network (who connects to whom). The NetLogo world is used only as a visualization space for agents and links.

Model Parameters:

- **β (BETA-IMBALANCE) $\in [0, 2]$** : Penalty coefficient for trait imbalance between warmth and competence
 - $\beta = 0 \rightarrow$ no penalty (only the product $W \times C$ matters)
 - $\beta = 2 \rightarrow$ strong penalty (strongly favors balanced profiles)
 - This parameter represents the "preference for balance" in social evaluation
- **τ (TAU-BASE-CONNECT) $\in [0, 0.3]$** : Baseline connection probability, independent of traits
 - $\tau = 0 \rightarrow$ connections based purely on attraction score
 - $\tau = 0.3 \rightarrow$ 30% baseline chance of connecting with anyone
 - This parameter represents contextual "social openness"

Exogenous factors:

The exogenous factors driving the model are:

1. Agents' preference for balanced warmth-competence profiles (modulated by β)
2. Contextual social openness (represented by τ)
3. Emotional responses to different trait combinations

These factors drive the formation of networks (or lack thereof) and hierarchization.

c. Process Overview and Scheduling

Temporality: Each time step represents a round of first impressions. Each agent perceives all other agents and chooses whom to connect with based on first impression evaluation.

At each time step (tick):

1. Perception

- Agent i observes every other agent j
- Perceived warmth and competence include small uniform noise: $\text{random_float}(0.05) - 0.025$
- Perceived values are clamped to $[0, 1]$

2. Emotional Evaluation

- For each j , agent i assigns an emotional category based on perceived traits:
 - **Admiration:** $W_j \geq 0.5 \text{ AND } C_j \geq 0.5$
 - **Pity:** $W_j \geq 0.5 \text{ AND } C_j < 0.5$
 - **Envy/Competition:** $W_j < 0.5 \text{ AND } C_j \geq 0.5$
 - **Contempt:** $W_j < 0.5 \text{ AND } C_j < 0.5$

3. Attraction Score Calculation

- Base score = $(W_j \times C_j) - (\beta \times |W_j - C_j|)$
- Emotional modifier applied based on category and agent i 's own traits
- Final attraction score = base score + emotional modifier

4. Candidate Selection

- Agent i ranks all others by attraction score (descending)
- Selects top 5 candidates (MAX-OUTGOING-LINKS)

5. Tie Formation

- For each candidate in top 5:

- Connection probability = $\min(1.0, \max(0.0, \text{attraction_score} + \tau))$
 - If random-float < probability AND link doesn't exist AND i hasn't reached max links:
 - Create link $i \rightarrow j$
 - Existing links remain (accumulation)
6. **Observation**
- Network metrics (degree, isolates, emerging clusters) are recorded

Order of execution: Perception → Evaluation → Attraction calculation → Candidate selection → Tie formation → Observation

2. Design Concepts

a. Theoretical Background

General concepts underlying the model:

The model is based on **social cognition theories** and **non-verbal communication theories**, particularly the **warmth-competence framework** (Fiske et al., 2007), which posits that warmth and competence are the two fundamental dimensions in first impressions.

From an **evolutionary perspective**, when interacting with someone for the first time, humans rapidly assess: (1) intent (warmth) - is this person a friend or foe? and (2) capability (competence) - can they act on their intentions? Non-verbal cues convey these dimensions and impact perceived attractiveness for social connection.

Deviation from initial formulation:

During implementation, we refined the attraction score calculation to better capture the theoretical construct of social preference. Instead of a simple additive formula ($W + C$), we use a **multiplicative base ($W \times C$) with an imbalance penalty ($\beta \times |W - C|$)**. This formulation:

1. Aligns better with social cognition theory, where warmth and competence **interact** rather than simply add up
2. Allows us to test whether balanced profiles (high on both dimensions) are more attractive than specialized profiles (high on one dimension only)
3. Introduces β as a theoretically meaningful parameter representing the "preference for balance" in social evaluation

Additionally, we introduced τ (TAU-BASE-CONNECT) to represent contextual "social openness," allowing us to explore how context moderates trait-based attraction.

b. Emergence

Network clusters, highly connected hubs, and isolated agents emerge from repeated local decisions. No agent has global knowledge of the overall structure. The final network topology is an emergent property of individual attraction evaluations.

c. Adaptation

None. Agents apply the same evaluation rules at every tick. There is no learning or strategy change.

d. Sensing

Agents sense the warmth and competence cues of all other agents, with slight perceptual noise (± 0.025) to simulate imperfect first impressions and create non-deterministic behavior.

e. Interaction

Interaction consists of perception and tie formation. There is no verbal communication or behavioral feedback between agents.

f. Stochasticity

- Random initialization of warmth and competence traits (uniform $[0,1]$)
- Noise in perception (uniform ± 0.025)
- Probabilistic tie formation based on attraction scores
- Random ordering of agent actions

g. Observation

The final network structure is observed after 30 ticks (or earlier if equilibrium is reached). Key metrics include:

- Degree distribution
- Network centrality measures (max, mean)
- Number of isolated agents
- Total link count
- Network density

3. Details

a. Initialization

1. Choose number of agents N (typically $N = 60$)
2. Create N agents with:
 - random warmth _{i} $\in [0,1]$ (uniform distribution)
 - random competence _{i} $\in [0,1]$ (uniform distribution)
 - num-links _{i} = 0
3. Assign visual appearance:
 - Blue: high W , high C (admiration/charisma)
 - Red: high W , low C (pity)
 - Yellow: low W , high C (envy/competition)
 - Gray: low W , low C (contempt/neglect)
4. Place agents in a circle in the environment (for visualization)

5. Remove all links
6. Set tick = 0

Table 2: Expected distribution of agents across emotional categories under uniform random initialization. Actual observed proportions may vary due to stochastic trait assignment.

Table 2: Theoretical Distribution of Emotional Categories

Agents generated randomly with uniform distribution in $[0,1]$ for warmth and competence

Category	W (Warmth)	C (Competence)	Expected Proportion	NetLogo Color
Admiration	≥ 0.5	≥ 0.5	25%	Blue
Pity	≥ 0.5	< 0.5	25%	Red
Envy	< 0.5	≥ 0.5	25%	Yellow
Contempt	< 0.5	< 0.5	25%	Gray

Note: With uniform random initialization, each quadrant should theoretically contain approximately 25% of agents. Actual proportions vary by simulation run.

b. Input Data

The model does not use any external empirical data. All agent attributes are generated internally from specified distributions.

c. Submodels

Submodel 1: Perceived Attraction Score

For agent i evaluating agent j :

Step 1: Noisy perception

$\text{noise_w} \sim \text{Uniform}(-0.025, +0.025)$

$\text{noise_c} \sim \text{Uniform}(-0.025, +0.025)$

$W'_j = \text{clamp}(W_j + \text{noise_w}, 0, 1)$

$C'_j = \text{clamp}(C_j + \text{noise_c}, 0, 1)$

Step 2: Base score with imbalance penalty

$\text{base_score} = (W'_j \times C'_j) - (\beta \times |W'_j - C'_j|)$

Step 3: Emotional modifier

Determine emotional category based on W'_j and C'_j :

IF $W'_j \geq 0.5$ AND $C'_j \geq 0.5$: [Admiration]

$\text{emotional_mod} = 0.0$

IF $W'_j \geq 0.5$ AND $C'_j < 0.5$: [Pity]

IF $W_i \geq 0.5$:

emotional_mod = +0.2 (warm agents sympathize)

ELSE:

emotional_mod = 0.0

IF $W'_j < 0.5$ AND $C'_j \geq 0.5$: [Envy/Competition]

IF $C_i \geq 0.5$:

emotional_mod = -0.2 (competent agents feel competitive)

ELSE:

emotional_mod = 0.0

IF $W'_j < 0.5$ AND $C'_j < 0.5$: [Contempt]

emotional_mod = -0.5 (strong avoidance)

Step 4: Final attraction score

attraction_score_ij = base_score + emotional_mod

Rationale:

- The multiplicative term ($W \times C$) ensures agents high on both dimensions are most attractive
- The imbalance penalty ($\beta \times |W - C|$) reduces attraction for unbalanced profiles
- Emotional modifiers capture homophily effects (warm agents prefer warm targets) and competition (competent agents avoid competent rivals)

Submodel 2: Connection Probability

$p_{\text{connect_ij}} = \min(1.0, \max(0.0, \text{attraction_score_ij} + \tau))$

Where:

- attraction_score_ij: from Submodel 1
- τ (TAU-BASE-CONNECT): baseline connection probability [0, 0.3]
- The min/max ensures probability stays in [0, 1]

Rationale: τ represents contextual factors (e.g., a highly social event vs. a formal meeting) that affect baseline willingness to connect, independent of individual traits.

Submodel 3: Tie Formation Process

For each agent i at each tick:

1. Calculate `attraction_score` for all other agents j
2. Sort agents by `attraction_score` (descending)
3. Select TOP 5 candidates (or fewer if less than 5 others exist)
4. For each candidate in TOP 5:
IF (current outgoing links < MAX-OUTGOING-LINKS) AND
(link $i \rightarrow j$ does not exist):

$$p = \text{attraction_score_j} + \tau$$
$$p = \text{clamp}(p, 0, 1)$$

IF $\text{random-uniform}(0,1) < p$:
CREATE link $i \rightarrow j$

5. Stop when MAX-OUTGOING-LINKS reached OR all top candidates processed

Rationale:

- Agents have limited attention capacity (max 5 connections)
- They prioritize the most attractive targets
- Probabilistic formation allows for variability across simulation runs

Submodel 4: Visual Layout (for display only)

A spring-layout algorithm (layout-spring) adjusts agent positions for visualization. This has no effect on the model dynamics and can be disabled for performance during batch experiments.

Results: Factorial Experiment Findings :

Analysis 1: Network Connectivity and Parameter Effects

The results of the Factorial Experiment (FE) demonstrate a pattern that is both coherent and highly meaningful, directly supporting the model's core theoretical objective: understanding how the balance between warmth (W) and competence (C) shapes first-impression dynamics and, consequently, the structure of the emerging social network.

Average of count links	Column Labels					
Row Labels	0	0.1	0.15	0.2	0.3	Grand Total
0	157.3666667	158.4333333	157.9333333	160.9666667	156.0333333	158.1466667
0.02	156.4	159.2333333	159.3333333	158.6	160.6	158.8333333
0.04	155.3	162.6333333	160.3333333	157.6333333	159.9	159.16
0.06	150.4666667	153.1666667	161.4333333	158.1666667	157.2666667	156.1
0.08	148.3	161.7	159.3666667	166.7666667	160.1666667	159.26
0.1	148.7333333	155.6333333	158.3333333	159.1666667	161.1666667	156.6066667
0.12	149.0333333	159.4333333	159.1333333	161.4	161.3666667	158.0733333
0.14	149.2	158.6333333	159.6666667	158.1333333	158.9666667	156.92
0.16	141.2333333	155.2333333	157.2	159.2333333	160.7333333	154.7266667
0.18	139.5666667	154.5	161.5666667	159.3333333	160.1666667	155.0266667
0.2	139.9	154.3333333	159.5333333	161.3	160.9333333	155.2
Grand Total	148.6818182	157.5393939	159.4393939	160.0636364	159.7545455	157.0957576

Figure 1. Average link count across parameter configurations (7 TAU values $\times$ 11 BETA values)

1. The Macro-Level Influence of Context

Higher τ consistently raises network density. τ functions as environmental openness: it shows that network structure depends not only on individual W/C traits but also on how easily connections can form. Even when other factors limit ties, high τ partly compensates by keeping opportunities for connection alive.

2. The Micro-Level Power of the Mechanism

β controls how strongly W/C imbalance is penalized. As β increases, imbalanced agents become far less attractive, making the network sparser and pushing these agents to the periphery. Ties cluster around balanced profiles.

Conclusion: balanced W/C increases social desirability and centrality; imbalance is strongly punished.

3. Synthesis: Context Mitigates, But Does Not Eliminate, Selectivity

τ and β interact in a clear macro–micro dynamic: τ sets how easy it is to connect; β determines who gets chosen. High τ creates denser networks, but β still differentiates sharply, high- β conditions produce more inequality and clustering even in open environments.

Overall, openness reduces but cannot erase the disadvantages of W/C imbalance. Balanced profiles rise to the center, while τ shapes the overall breadth and robustness of the emerging structure.

Analysis 2: Social Exclusion Dynamics

The second analytical goal of the FE was to identify the conditions under which agents become fully socially excluded—ending the simulation with zero ties. The results show a clear,

coherent pattern driven by the interaction between the micro-level trait penalty (β) and the macro-level relational openness (τ).

Average of count turtles with [count link-neighbors = 0] Column Labels						
Row Labels	0	0.1	0.15	0.2	0.3	Grand Total
0	0.766666667	0.866666667	1	0.533333333	1.033333333	0.84
0.02	0.9	0.666666667	0.733333333	0.6	0.833333333	0.746666667
0.04	1.166666667	0.733333333	0.6	0.9	0.666666667	0.813333333
0.06	1.833333333	1.133333333	0.5	1	1.033333333	1.1
0.08	1.733333333	0.666666667	0.8	0.3	0.7	0.84
0.1	2.2	1.033333333	0.866666667	0.733333333	0.633333333	1.093333333
0.12	2.066666667	0.566666667	1.033333333	0.666666667	0.833333333	1.033333333
0.14	2.166666667	0.733333333	0.566666667	0.866666667	0.633333333	0.993333333
0.16	3.366666667	1.233333333	0.4	0.5	0.566666667	1.213333333
0.18	2.866666667	0.966666667	0.633333333	0.466666667	0.6	1.106666667
0.2	3.133333333	1.133333333	0.8	0.9	0.566666667	1.306666667
Grand Total	2.018181818	0.884848485	0.721212121	0.678787879	0.736363636	1.007878788

Figure 2. Average number of isolated agents across configurations. Higher BETA and lower TAU increase exclusion

1. The Causal Role of Trait Imbalance (β)

Rising β sharply increases the number of isolated agents. The penalty on warmth–competence imbalance makes uneven profiles far less socially desirable, causing many to fail to form even one tie.

Strong imbalance, when penalized, can push agents from low centrality to complete exclusion.

2. The Protective Role of Context (τ)

Higher τ (more relational openness) protects agents from exclusion. In open environments, even imbalanced profiles can form ties; when τ is low, the same imbalance triggers much harsher exclusion.

The effect is that low τ amplifies β ; high τ buffers it.

3. Emergence of Structured Isolation

Exclusion emerges from the interplay between:

$\beta \rightarrow$ who is considered socially desirable

$\tau \rightarrow$ how forgiving the environment is

The strongest isolation arises when imbalanced traits meet a low-openness context. Balanced profiles or open environments reduce this risk, but the mechanism shows how simple non-verbal perceptions can scale into persistent social inequalities.

Analysis 3: Hub Formation and Centrality

Contrary to the initial theoretical expectation that the model would produce a clear, strong hierarchical differentiation (i.e., "superstar hubs"), the results reveal a remarkably stable and flat network structure in terms of maximum centrality. Across all parameter combinations tested, the most connected agent in the network consistently accumulated a moderate number of incoming links, clustering tightly around 10 ties (ranging only from approximately 9.5 to 10.5). This stability points to key limitations and constraints within the modeled first-impression process.

Average of max [count link-neighbors] of turtles	Column Labels					
Row Labels	0	0.1	0.15	0.2	0.3	Grand Total
0	10.3	10.13333333	10.46666667	10.63333333	10.16666667	10.34
0.02	10.66666667	10.56666667	10.46666667	11.1	10.53333333	10.66666667
0.04	10.53333333	10.73333333	10.26666667	10.66666667	10.5	10.54
0.06	10.16666667	10.06666667	10.36666667	10.23333333	10.36666667	10.24
0.08	9.8	10.33333333	10.6	10.66666667	11	10.48
0.1	9.63333333	10.03333333	10.06666667	10.36666667	10.63333333	10.14666667
0.12	10.06666667	10.2	10.43333333	10.63333333	10.73333333	10.41333333
0.14	9.66666667	10.33333333	10.26666667	10.56666667	10.46666667	10.26
0.16	9.56666667	10.2	10.56666667	10.33333333	10.7	10.27333333
0.18	9.36666667	9.93333333	10.63333333	10.36666667	10.36666667	10.13333333
0.2	9.63333333	10.46666667	10.3	10.2	10.2	10.16
Grand Total	9.945454545	10.27272727	10.4030303	10.52424242	10.51515152	10.33212121

Figure 3. Maximum in-degree centrality across configurations.

1. The Null Effect of the Micro-Mechanism (β)

β has almost no impact on maximum in-degree.

Observation: Even strong imbalance penalties do not reduce the top hub's centrality.

Interpretation: β shapes who gets chosen, but not how high the peak centrality can go. A small set of perfectly balanced agents consistently absorbs ties up to the system's structural limit.

2. The Weak Contextual Effect of Openness (τ)

τ shows only a slight positive effect on maximum degree ($\approx 9.95 \rightarrow 10.52$).

Interpretation: Greater openness makes it marginally easier for already-attractive agents to gain extra ties, but the effect is small and does not create extreme inequality.

Overall, we have a ceiling effect and moderate centrality. No "superstar hubs" emerge. The model consistently produces moderate hubs, indicating that first-impression dynamics alone cannot generate extreme hierarchy. And early network formation driven solely by non-verbal first impressions yields moderately central actors within a relatively flat hierarchy, falling short of producing the strong inequalities originally expected.

Analysis 4: Emotional Categories and Network Position

Table 1. Contextual configurations selected for manual simulation analysis. Each context tests specific theoretical predictions about hierarchy and exclusion

Table 1: Summary of Three Contextual Configurations

Config	τ (TAU)	β (BETA)	Context Represented	Research Question
1. Baseline	0.15	1.0	Typical/Neutral	General pattern: Who dominates? Who is isolated?
2. Restrictive	0.1	2.0	Competitive/Selective	Maximum exclusion? Extreme hierarchy?
3. Permissive	0.3	0.5	Inclusive/Cooperative	Compensation possible? Reduction of exclusion?

Note: These configurations test how contextual parameters (social openness and imbalance penalty) shape network outcomes.

Table 3. Parameter values for the three manually examined simulation contexts. Values represent contrasting levels of social openness (τ) and preference for balanced profiles (β).

Table 3: Parameter Values for Three Manual Simulation Contexts

Context	τ (Social Openness)	β (Imbalance Penalty)	Characteristics
Baseline	0.15	0.10	Moderate openness, Low penalty
Permissive	0.30	0.05	High openness, Minimal penalty
Restrictive	0.05	0.20	Low openness, High penalty

Note: These values align with the tested parameter ranges: $\tau \in [0, 0.3]$, $\beta \in [0, 0.2]$

Contrary to the initial theoretical expectation that the model would produce a clear, strong hierarchical differentiation (i.e., "superstar hubs"), the results reveal a remarkably stable and flat network structure in terms of maximum centrality. Across all parameter combinations tested, the most connected agent in the network consistently accumulated a moderate number of incoming links, clustering tightly around 10 ties (ranging only from approximately 9.5 to 10.5). This stability points to key limitations and constraints within the modelled first-impression process.

1. The Null Effect of the Micro-Mechanism (β)

β has almost no impact on maximum in-degree.

Observation: The top hub's centrality stays nearly constant, regardless of how strong the imbalance penalty is.

Interpretation: β changes who gets chosen, but not how high centrality can rise. A small group of perfectly balanced agents consistently absorbs ties up to the model's structural cap.

2. The Weak Contextual Effect of Openness (τ)

τ shows only a slight positive effect on hub centrality ($\approx 9.95 \rightarrow 10.52$).

Interpretation: Greater openness gives already-attractive agents a slightly higher chance of gaining extra ties, but the effect is weak and does not create extreme hierarchy.

Manual Examination of Contextual Configurations

A closer review of three specific parameter settings shows how the four Warmth–Competence emotional categories shape average network position.

Table 4: Average Connectivity by Emotional Category Across Contexts

Context	Admiration (W+, C+)	Pity (W+, C-)	Envy (W-, C+)	Contempt (W-, C-)
Baseline	5.89 links	7.29 links	5.00 links	3.92 links
Permissive	6.15 links	6.85 links	5.06 links	2.54 links
Restrictive	6.13 links	6.41 links	3.83 links (8.3% isolated)	1.94 links (25% isolated)

Key Finding: In all contexts, **Pity agents** (high warmth, low competence) achieved equal or higher connectivity than Admiration agents, demonstrating **warmth primacy** in low-penalty environments.

Key Findings

1. Surprising Resilience Against Social Exclusion

Very few agents became isolated: none in the Baseline or Permissive contexts, and only a handful in the Restrictive one (1 Envy, 4 Contempt).

Interpretation: This contradicts the initial expectation that high imbalance penalties would generate systematic exclusion. The likely causes are the low β range (0.05–0.20), which creates weak penalties, and the buffering effect of τ , which provides enough openness over 30 ticks to prevent most agents from ending with zero ties.

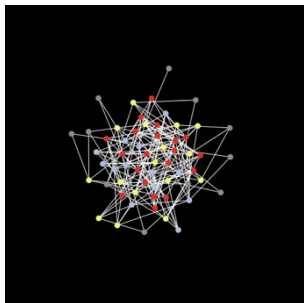


Image 1. Baseline

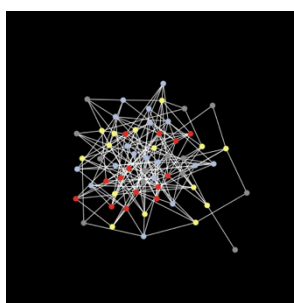


Image 2. Permissive

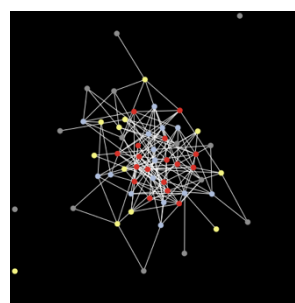


Image 3. Restrictive

2. Hierarchy Persists, Driven by Warmth Primacy

Even with low exclusion, a stable hierarchy emerged.

Observation: In Baseline and Permissive contexts, Pity agents (high warmth, low competence) had higher average connectivity than Admiration agents (high–high).

Interpretation: In low-penalty environments, warmth dominates tie formation—possibly reinforced by the model’s emotional modifier that favors warm targets.

3. The Dominance of Warmth Over Competence

Across all three tested configurations, agents categorized as **high in warmth** (Admiration and Pity) consistently outperformed those **low in warmth** (Envy and Contempt), regardless of their competence levels.

Table 5. Warmth primacy effect: Average connectivity by warmth dimension across contexts. High-warmth agents (Admiration + Pity) consistently outperform low-warmth agents (Envy + Contempt) regardless of competence levels.

Table 5: Warmth Dimension Dominance Analysis

Average connectivity aggregated by warmth level across all four emotional categories

Context	High Warmth (Admiration + Pity)	Low Warmth (Envy + Contempt)	Difference
Baseline	6.59 links	4.46 links	+2.13
Permissive	6.50 links	3.80 links	+2.70
Restrictive	6.27 links	2.89 links	+3.38

Interpretation: Agents high in warmth consistently outperform low-warmth agents by 2-3 links on average, with the gap widening in restrictive contexts. This confirms that warmth is more critical than competence for social integration in first-impression networks.

This supports evolutionary theories arguing that warmth (intent) is evaluated before competence (ability). The model demonstrates clear warmth primacy: losing warmth harms social integration more than losing competence.

Conclusion: The Ceiling Effect and Moderate Centrality

The most critical finding is the **absence of "superstar hubs"** (e.g., agents with 15–20+ links). Instead, the model consistently produces moderate hubs. This suggests that the pure first-impression process, as formalized here, is **insufficient to generate extreme social inequality**. Several factors likely contribute to this ceiling effect:

1. **Saturation Constraint:** The strongest constraint is imposed by the rule limiting agents to **a maximum of 5 outgoing links per tick**. This structural boundary caps the total number of ties that can be produced per simulation step, effectively preventing any single agent from accumulating an extreme number of ties within the 30-tick run.

2. **Homogeneous Environment:** The static and homogeneous nature of the W/C profiles means attractiveness differences do not escalate over time. Without feedback loops (e.g., preferential attachment where high-degree agents become even more attractive) or dynamic changes, the selection process quickly hits a saturation point among the most desirable targets.

In summary, the results indicate that the initial phase of network formation governed by fixed non-verbal perceptions leads to **mildly central individuals** within a relatively flat social hierarchy. The combination of structural constraints and the limitations of a static mechanism prevents the creation of the extreme hierarchical differentiation initially anticipated.

Conclusion

This agent-based model shows that simple evaluations based on non-verbal warmth and competence cues are sufficient to generate patterned social inequality in first-impression networks. Across 63 configurations (1650 runs), three mechanisms emerged:

Warmth primacy: High-warmth agents (Admiration/Pity) consistently formed more ties, supporting evolutionary accounts that intent is assessed first.

Context–mechanism interaction: The imbalance penalty (β) increased inequality and isolation, while baseline openness (τ) softened—but did not remove—these effects.

Moderate stratification: The model produced mild hubs, indicating that static first-impression processes alone cannot generate the strong inequalities seen in real networks without dynamic feedback.

This work strengthens middle-range theory by turning abstract constructs (warmth, competence, charisma) into a testable mechanism. It illustrates Coleman’s Boat: macro-context (τ) shapes micro-evaluations (W/C), which scale into macro-outcomes (network structure).

The parameter range used (β : 0–0.2; τ : 0–0.3) reflects relatively open contexts; future work should test extreme values ($\beta > 1$, $\tau < 0.05$). Adding learning, reputation, and longitudinal dynamics would increase realism and allow exploration of preferential attachment.

In sum, this study offers a minimal mechanistic account of how basic perceptual processes shape social structure, and provides a foundation for future theory-driven and empirical work.

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